

Connectivity Diagnostics in a Pilot Generative Surrogate of Zero-Degree Calorimeter Showers

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Abstract

We report an exploratory case study of one hierarchical generative surrogate for single-neutron Geant4 showers in a 6,790-channel Zero-Degree Calorimeter readout. The selected checkpoint was trained on 26,624 events and evaluated on a repeatedly inspected development bank of 10,000 conditions at 50–250 GeV. Coarse aggregation gives a favorable but incomplete picture: the generated and reference mean total deposits differ by 0.045 GeV (1.0%), while opposing ECAL and HCAL shifts partially cancel. On nonempty events, the mean number of active layers differs by 0.25, but the generated shower reaches 4.42 layers farther downstream on average and contains 3.95 more inactive layers within its occupied span. The mean number of active channels differs by -28.7 (-1.8%), whereas the strict-positive support contains 35.97 more weak components on the model graph and its mean largest-component fraction is lower by 6.03 percentage points. These are descriptive observations for one checkpoint, one graph, and a zero deposit threshold; event-level arrays needed for uncertainty estimates, threshold scans, and graph sensitivity studies were not retained. The comparison therefore identifies a connectivity discrepancy in this artifact rather than establishing general surrogate fidelity or a causal model failure.

1 Introduction

Detailed particle transport is central to detector design and high-energy-physics analysis, but calorimeter shower simulation is computationally expensive. Generative surrogates based on normalizing flows, diffusion, point clouds, graph networks, and conditional flow matching have consequently been developed for faster sample production [1–5]. Their evaluation must probe more than a single response summary. CaloChallenge compares one-dimensional observables, physics distances, classifier tests, model size, and generation time [6]; systematic studies of HEP generative metrics likewise show that different tests respond to different distributional failures [7].

Zero-Degree Calorimeters (ZDCs) are also the subject of dedicated surrogate studies. Work on the ALICE ZDC has considered variational autoencoders, generative adversarial networks, diffusion models, conditional flow matching, and mixtures of generative experts [8–11]. Those studies concern a different detector representation from the raw deposited-energy vector examined here, so their numerical results are not direct baselines for this checkpoint.

Spatial organization is physically relevant in calorimetry. For example, ATLAS topological clustering groups neighboring cells above signal-significance thresholds and uses the resulting cluster shapes in reconstruction [12]. This does not make every graph component a detector observable: connectivity depends on the readout mapping, the adjacency definition, and the energy threshold. It does motivate asking whether a surrogate that matches response and occupancy summaries also reproduces the organization of its occupied readout support.

We address that question for one pilot-trained checkpoint on an irregular 65-layer readout. The available evidence is aggregate and comes from a development bank that was inspected repeatedly. We therefore treat the study as a diagnostic case report. Its narrow result is that similar mean occupancies

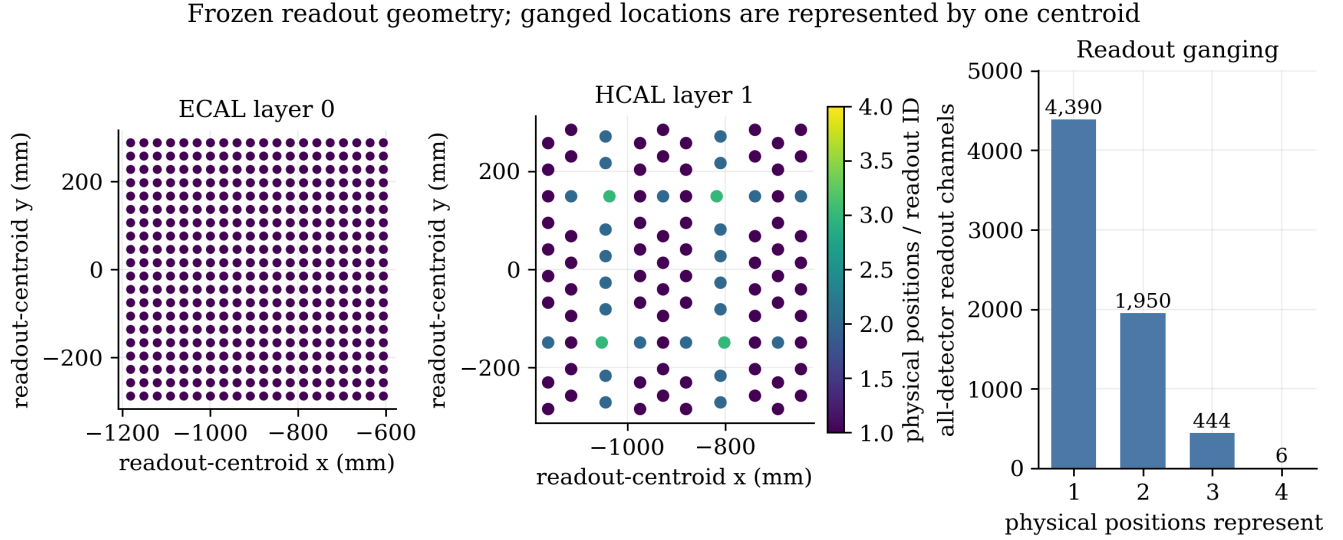


Figure 1: Frozen readout geometry. The left panels show ECAL layer 0 and HCAL layer 1. The ganging histogram covers all 6,790 readout channels; its first bar contains 400 ECAL and 3,990 HCAL channels.

coexist with a later longitudinal reach and a more disconnected strict-positive support under the graph used by the model. We do not infer statistical significance, performance across retraining, or the mechanism that produced the discrepancy.

2 Simulation target and evaluation population

For each event, the stored condition is an incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$. The model input is

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \mathbf{p}/\|\mathbf{p}\|, \quad (1)$$

where $K_{\text{inc}} = E - m_n$. The two energy features are deterministically related by $E = K_{\text{inc}} + m_n$; retaining both is a frozen implementation choice and does not add independent conditioning information.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ contains raw, nondigitized energy deposits summed by valid readout ID. It is not the incident-neutron energy or the total energy lost in all detector material. Layer 0 contains 400 electromagnetic-calorimeter (ECAL) channels. Hadronic-calorimeter (HCAL) layers 1–63 contain 100 channels each and layer 64 contains 90, for 6,390 HCAL channels. Of the HCAL readout IDs, 2,400 gang two to four stable physical positions. Each ganged channel is represented by the unweighted centroid of its positions (Fig. 1). The released metadata does not specify the full material composition, Geant4 version, physics list, production cuts, field configuration, or whether p^μ is recorded at generation or calorimeter entry. The observed few-GeV readout response can therefore be interpreted only as energy in the stored active-readout target, not as a calibrated calorimeter response fraction.

The graph connects every channel to its eight nearest same-layer centroids. For each node in layer $\ell = 0, \dots, 63$, four nearest centroids are selected in layer $\ell + 1$, and each selected longitudinal pair is stored in both directions. This gives $8 \times 6790 = 54,320$ lateral directed edges and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal directed edges, or 107,920 in total. The same graph is used by the generator and by the connectivity diagnostic. It is an implementation graph, not a uniquely established physical-neighbor graph, and centroiding removes separation within a ganged channel.

The preparation record contains 764,940 events over 0–300 GeV, divided by deterministic event hash into 612,482 training, 76,158 validation, and 76,300 nominal test events. The checkpoint studied here used a smaller pilot of 26,624 training events and 6,656 validation events. Its training population is 4.35% of the canonical training partition.

any one term caused the observed structure. AdamW optimization used a peak learning rate of 3×10^{-4} , batch size six, accumulation over four batches, gradient clipping at one, and training seed 20260723. Epoch 90 has the lowest recorded validation objective among 104 rows spanning epochs 11–114 and defines the single checkpoint analyzed below. Validation objectives are lower than training objectives over much of this record; the retained evidence does not identify the cause, so we do not interpret that ordering.

4 Diagnostics

We first compare all-event mean total, ECAL, and HCAL deposits. Eight 25 GeV incident-energy bins are also available as aggregate means and standard deviations. They are used only to identify the range of descriptive deviations because the retained report contains no binwise uncertainty estimates. We omit the pooled Wasserstein distance and the historical reference-half comparison: the sample sizes and condition distributions differ, and the stored bootstrap does not provide a calibrated uncertainty statement for that comparison.

A layer or channel is active when its stored energy is strictly positive; no detector or analysis threshold is applied. For a nonempty event with first and last active layers F and L , the occupied span is $L - F + 1$, and the number of inactive layers inside it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_{\ell}. \quad (4)$$

This counts inactive layers, not contiguous runs. A shower has an interior gap when $G > 0$. On the undirected form of the model graph, the active support S separates into weak components $C_1, \dots, C_m$; we report m and $\max_j |C_j|/|S|$. All support summaries are conditioned on nonempty events. The reference and generated nonempty sample sizes are 9,907 and 9,858, respectively.

The retained condition-only pipeline control has AUROC 0.500. Because it contains no shower variables, it checks only that identical condition distributions are presented to the two labels; it is not a two-sample test of the generated showers. A valid classifier test would require held-out discrimination [7] and, for this paired design, a split that keeps each matched reference–generated pair together.

5 Results

Table 1 reports absolute differences so that ratios on small denominators do not exaggerate the comparison. The all-event mean total deposit differs by 0.045 GeV, but this sum combines an ECAL excess of 0.175 GeV with an HCAL deficit of 0.129 GeV. Across the eight incident-energy bins, the generated-minus-reference mean difference ranges from -7.7% to $+7.7\%$, while the standard-deviation difference ranges from -10.9% to $+9.6\%$. Without binwise uncertainties these ranges diagnose possible cancellation but do not establish a statistically resolved energy dependence.

The longitudinal discrepancy is more directly described as a shift in reach than as a ratio of gap counts. The generated shower has nearly the same mean number of active layers, but its first active layer is 0.225 layer later and its last active layer is 4.42 layers later. The occupied span consequently grows by 4.20 layers while the active-layer count grows by only 0.25, leaving 3.95 additional inactive layers inside the span. Figure 3 shows the corresponding mean energy allocation.

At channel level, the generated events contain 28.7 fewer active channels on average but 35.97 more weak components. Their largest component contains 6.03 percentage points less of the active support. Under the declared graph and strict-positive definition, the support is therefore more disconnected even though its mean cardinality is similar. The available aggregates do not record component energies. They cannot determine whether the additional components carry substantial energy or arise from small positive deposits produced by the selected-channel softmax.

Table 1: Aggregate development-bank results. Deposit rows and zero-deposit counts use all 10,000 pairs. Support rows are conditional on nonempty events (9,907 reference and 9,858 generated). Percentage-point differences are denoted pp. No uncertainty intervals can be reconstructed from the retained aggregates.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

There are 49 more zero-deposit generated events than reference events. A significance calculation would require the joint paired zero/nonzero counts, which were not retained; the marginal counts alone do not support one. The result is reported because the visibility hurdle is part of the sampled model and because excluding empty events changes the denominators of every support statistic.

The decoder-level checks passed for all 1,250 batches of eight generated events. No nonfinite value, negative value, invalid support, requested-versus-realized count mismatch, or positive-support mismatch was recorded. Maximum event and layer budget residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV under the evaluator’s batch-wide absolute-plus-relative tolerance. These checks establish consistency with sampled budgets and counts for this generated bank; they do not test agreement with Geant4.

6 Discussion

The empirical statement supported by these aggregates is specific: one pilot checkpoint produces a later and more disconnected strict-positive readout support than its paired Geant4 sample while retaining similar mean layer and channel multiplicities. Exact cardinality and budget allocation leave the spatial arrangement underdetermined, so this coexistence is compatible with the decoder construction. The comparison does not show that the factorization caused the discrepancy. Limited training data, checkpoint selection, numerical integration, the shared graph, and the zero threshold remain alternative explanations.

Connectivity is one useful projection of shower structure, but it is not a complete or representation-independent fidelity metric. CaloChallenge and CaloDREAM illustrate the broader practice of combining one-dimensional shower observables with multivariate diagnostics [5, 6]. The HEP metric literature similarly recommends complementary tests rather than a single scalar ranking [7]. For this readout, a defensible connectivity result would additionally require thresholds tied to detector noise or digitization, energy-weighted component summaries, alternative adjacency graphs, and distributions resolved in incident energy and direction.

The missing tests have distinct purposes. Repeating generation with several sampling seeds would measure Monte Carlo variability for the fixed checkpoint. Neighboring checkpoints and independent training seeds would test selection and training variability. A 26k–100k–full-data scaling study would separate pilot undertraining from a persistent model limitation. Replacing generated upstream variables with their reference counterparts in an oracle cascade would localize where the discrepancy enters the hierarchy. Solver-step scans would test numerical integration, while sequential layer activity and connectivity-aware

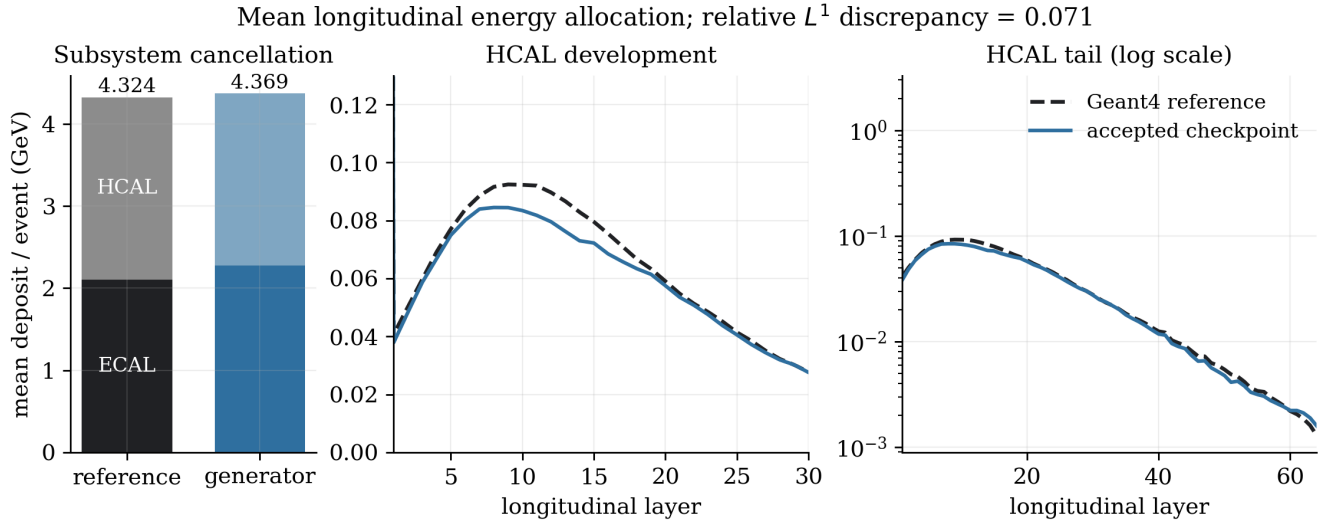


Figure 3: Mean longitudinal allocation on D_{dev} . The stacked bars expose the opposing ECAL and HCAL shifts. The layer panels show HCAL layers only; ECAL layer 0 is isolated in the bars. The curves are descriptive means without event-level uncertainty bands.

support would test specific model alternatives. None of these inferences can be recovered by further analysis of the present aggregate file.

The development status is also decisive. D_{dev} was repeatedly inspected, its overlap with pilot validation is unresolved, and event-level arrays were not retained. A physics-performance paper would require a locked evaluation bank with documented identities, paired uncertainty estimates, full structural distributions, a valid shower-aware classifier test, reconstructed-observable comparisons, and end-to-end timing. It would also require a complete reference-simulation specification and the applicable collaboration approval for release. Until those conditions are met, the present manuscript is an exploratory computational-physics case study rather than a validated fast-simulation result.

7 Conclusion

For the single pilot checkpoint studied here, close mean occupancies do not coincide with close readout-support organization. Relative to the paired Geant4 sample, the generated events contain 0.25 more active layers but reach 4.42 layers farther downstream, leaving 3.95 more inactive layers within the occupied span. They contain 28.7 fewer active channels but 35.97 more weak components on the model graph, with a 6.03-percentage-point reduction in the largest-component fraction. Opposing ECAL and HCAL shifts also partially cancel in the mean total deposit.

These values describe one repeatedly inspected development bank under a strict-positive threshold and one shared graph. They establish a reproducible discrepancy in this artifact. They do not establish its statistical significance, physical relevance after digitization, persistence across model fits, or causal origin. Those questions require the event-level, threshold, graph, seed, and locked-evaluation studies specified above.

Code and data availability

The manuscript source, aggregate validation report, event-independent geometry summary, training-history extract, figure builder, and QA records are available at <https://github.com/JulianAttemptsCoding/fast-mc-paper>. The collaboration-owned event file and checkpoint are not redistributed. The public

package reproduces the manuscript and aggregate arithmetic; it cannot retrain the model, regenerate showers, or reconstruct event-level tests.

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Author contributions

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

Use of generative AI

Generative-AI tools were used for editorial assistance, code generation, literature discovery, and adversarial manuscript auditing. Numerical statements were checked against versioned project artifacts during this revision; the author retains responsibility for verifying and approving the final manuscript.

References

- [1] C. Krause and D. Shih. “Fast and Accurate Simulations of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 107 (2023), p. 113003. DOI: [10.1103/PhysRevD.107.113003](https://doi.org/10.1103/PhysRevD.107.113003). arXiv: [2106.05285](https://arxiv.org/abs/2106.05285).
- [2] O. Amram and K. Pedro. “Denoising Diffusion Models with Geometry Adaptation for High Fidelity Calorimeter Simulation”. In: *Physical Review D* 108 (2023), p. 072014. DOI: [10.1103/PhysRevD.108.072014](https://doi.org/10.1103/PhysRevD.108.072014). arXiv: [2308.03876](https://arxiv.org/abs/2308.03876).
- [3] D. Kobylanski, N. Soybelman, E. Dreyer, and E. Gross. “Graph-Based Diffusion Model for Fast Shower Generation in Calorimeters with Irregular Geometry”. In: *Physical Review D* 110 (2024), p. 072003. DOI: [10.1103/PhysRevD.110.072003](https://doi.org/10.1103/PhysRevD.110.072003). arXiv: [2402.11575](https://arxiv.org/abs/2402.11575).
- [4] E. Buhmann et al. “CaloClouds II: Ultra-Fast Geometry-Independent Highly-Granular Calorimeter Simulation”. In: *Journal of Instrumentation* 19.04 (2024), P04020. DOI: [10.1088/1748-0221/19/04/P04020](https://doi.org/10.1088/1748-0221/19/04/P04020). arXiv: [2309.05704](https://arxiv.org/abs/2309.05704) [[physics.ins-det](#)].
- [5] L. Favaro, A. Ore, S. Palacios Schweitzer, and T. Plehn. “CaloDREAM: Detector Response Emulation via Attentive Flow Matching”. In: *SciPost Physics* 18 (2025), p. 088. DOI: [10.21468/SciPostPhys.18.3.088](https://doi.org/10.21468/SciPostPhys.18.3.088). arXiv: [2405.09629](https://arxiv.org/abs/2405.09629).
- [6] C. Krause et al. “CaloChallenge 2022: A Community Challenge for Fast Calorimeter Simulation”. In: *arXiv preprint* (2024). arXiv: [2410.21611](https://arxiv.org/abs/2410.21611) [[physics.ins-det](#)].
- [7] R. Kansal et al. “Evaluating Generative Models in High Energy Physics”. In: *Physical Review D* 107 (2023), p. 076017. DOI: [10.1103/PhysRevD.107.076017](https://doi.org/10.1103/PhysRevD.107.076017). arXiv: [2211.10295](https://arxiv.org/abs/2211.10295) [[hep-ex](#)].
- [8] J. Dubiński, K. Deja, S. Wenzel, P. Rokita, and T. Trzciński. “Machine Learning Methods for Simulating Particle Response in the Zero Degree Calorimeter at the ALICE Experiment, CERN”. In: *AIP Conference Proceedings* 3061 (2024), p. 040001. DOI: [10.1063/5.0203567](https://doi.org/10.1063/5.0203567). arXiv: [2306.13606](https://arxiv.org/abs/2306.13606) [[cs.CV](#)].
- [9] M. Kita, J. Dubiński, P. Rokita, and K. Deja. *Generative Diffusion Models for Fast Simulations of Particle Collisions at CERN*. 2024. arXiv: [2406.03233](https://arxiv.org/abs/2406.03233) [[physics.data-an](#)].
- [10] M. Wojnar. *Even Faster Simulations with Flow Matching: A Study of Zero Degree Calorimeter Responses*. 2025. arXiv: [2507.18811](https://arxiv.org/abs/2507.18811).
- [11] P. Będkowski, J. Dubiński, F. Szatkowski, K. Deja, P. Rokita, and T. Trzciński. *ExpertSim: Fast Particle Detector Simulation Using Mixture-of-Generative-Experts*. 2025. arXiv: [2508.20991](https://arxiv.org/abs/2508.20991) [[cs.CV](#)].
- [12] ATLAS Collaboration. “Topological Cell Clustering in the ATLAS Calorimeters and Its Performance in LHC Run 1”. In: *European Physical Journal C* 77 (2017), p. 490. DOI: [10.1140/epjc/s10052-017-5004-5](https://doi.org/10.1140/epjc/s10052-017-5004-5). arXiv: [1603.02934](https://arxiv.org/abs/1603.02934) [[hep-ex](#)].
- [13] W. Kool, H. van Hoof, and M. Welling. “Stochastic Beams and Where to Find Them: The Gumbel-Top- k Trick for Sampling Sequences Without Replacement”. In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 3499–3508.